Indiana University
Fall 2020

SPEA-E 552
A. Ward

Name: SOLUTION + GRADE NOTES

Quiz #5 (Lectures 19-22)

$\rho_{GAS} = 1.2\ kg/m^3$

1. Consider the use of a settling chamber to remove particulates from a gas stream. There is a waste gas stream of 10 m³/s and a settling chamber currently exist that is 5-m wide, 5-m tall, and 10-m long. Particles in the gas stream have a density of 1,500 kg/m3 and the gas stream has a viscosity of 2x10⁻⁵ kg/(m*s).

A. What diameter particle has a 95% removal in the current design?

+1 APPROACH
+1 #
+1 UNITS

GAS VELOCITY $= \frac{Q}{LW} = \frac{10\ m^3/s}{5m \times 5m} = \cancel{0.2\ m/s}\ 0.4\ \frac{m}{s}$

+1 APPROACH
+1 UNITS
+1 #

$$E = \frac{9.81\ m/s^2 \times (1500\ kg/m^3 - 1.2\ kg/m^3)(D)^2}{18(2\times10^{-5}\ kg/m\cdot s)(0.4\ \frac{m}{s})} = 0.95 \longrightarrow \boxed{D = 0.000096\ m = 96\ \mu m}$$

B. What is the removal efficiency of this settling chamber for particles of 50 um in diameter?

SAME

$$E = \frac{9.81\ m/s^2 \times (1500\ kg/m^3 - 1.2\ kg/m^3)(0.00005\ m)^2}{18 \times 2\times10^{-5}\ kg/m\cdot s \times 0.4\ \frac{m}{s}} = \boxed{25.5\ \%\ \text{OR}\ 0.255}$$

C. If a removal efficiency of 90% is desired for 50 um diameter particles, what height of settling chamber would be required?

SAME

$$0.90 = \frac{9.81\ m/s^2 \times (1500\ kg/m^3 - 1.2\ kg/m^3) \times (0.00005\ m)^2}{18 \times 2\times10^{-5}\ kg/m\cdot s \times V_g}$$

$$\longrightarrow V_g = 0.11\ \frac{m}{s}$$

SAME

$$0.11\ \frac{m}{s} = \frac{10\ m^3/s}{5m \times H} \longrightarrow \boxed{H = 18.2\ m}$$

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2. A cyclone is used as a scalping unit on a coal fired boiler. The height of the cyclone inlet is 0.75 m, the width of the cyclone inlet is 0.6 m, the length of the upper chamber is 2.0 m, and the length of the lower chamber is 5.0 m. The inlet gas velocity is 20.0 m/s and the particle density is 2,000 kg/m3. Assume gas viscosity of 1.85×10^{-5} kg/(m·s).

A. What is the removal efficiency of the cycle for particles of 100 um in diameter?

+1 APPROACH
+1 CALC

$$\# \text{ OF TURNS} = N = \frac{1}{0.75m}\left(2m + \frac{5.0m}{2}\right) = 6$$

+1 APPROACH
+1 #
+1 UNITS

$$E = \frac{\pi(6)(2000\,kg/m^3)(8\times10^{-5}m)^2(20\,m/s)}{9(1.85\times10^{-5}\,kg/m\cdot s)(0.6m)} = \boxed{0.483 \text{ OR } 48.3\%}$$

B. To have the largest increase in removal efficiency, which of the following scales should be doubled? (circle one of the following)

+1

A – Lower chamber length
B – Upper chamber length
C – entrance width
D – entrance height
E – 2 or more of the above are equivalent

A) $2\times L_B \Rightarrow E\uparrow$
B) $2\times L_C \Rightarrow E\uparrow$
C) $2\times W \Rightarrow E\downarrow$
D) $2\times H \Rightarrow E\downarrow$

C. Explain your logic for your response to Part B.

UP TO +4

$$E = \underbrace{\frac{1}{H}\left(L_B + \frac{L_C}{2}\right)}_{N}\frac{\pi \rho_p D_p^2 V_g}{9\mu W}$$

DOUBLE L_B BECAUSE THE IMPACT OF $2\times L_B > \frac{2\times L_C}{2}$

D. Calculate the removal efficiency for your selected response to Part B.

+1 APPROACH
+1 #
+1 UNITS

$$E = \frac{\pi(2000\,kg/m^3)(8\times10^{-5}m)^2(20\,m/s)}{9(1.85\times10^{-5}\,kg/m\cdot s)(0.6m)}\cdot\frac{1}{0.75m}\left(4m + \frac{5m}{2}\right)$$

$$= \boxed{69.8\% \text{ OR } 0.698}$$

3. In class we reviewed several options for resource recovery from wastewater. In terms of energy recovery, we specifically focused on (a) anaerobic digestion of wastewater sludge, and (b) the OmniProcessor.

If you were designing a new wastewater treatment infrastructure, what are two costs and two benefits of each system?

Anaerobic Digestor

UP TO 6 POINTS

Strengths:

1.

2.

Weaknesses:

1.

2.

ANSWERS WILL VARY

OmniProcessor

UP TO 6 POINTS

Strengths:

1.

2.

Weaknesses:

1.

2.

ANSWERS WILL VARY